SUPER Denoised Blueprint & Calibrated Quota Merging Pipeline

How MIRF Reliability Purification Restores Long-Tail Discovery under Bandwagon Shilling Attacks

1. Raw History

Observed Interactions

$(u, i_1), (u, i_2), \ldots$

Contains Injected

Bandwagon Shilling

(Attack Noise)

$\tilde{r}_{ui} \in \{1 \ldots 5\}$

2. MIRF Gating

Reliability Threshold

$w_{ui} \geq \theta_{\text{trust}}$

$(\theta_{\text{trust}} = 0.50)$

Shilling Pruned:

$w_{ui} \approx 0.05 \rightarrow$ **DROP**

Genuine Kept:

$w_{ui} \geq 0.85 \rightarrow$ **KEEP**

3. Clean Blueprint

Purified History

$B_u = \{i \mid w_{ui} \geq \theta\}$

Genuine User

Interests Restored

Tail Inclination:

$$\alpha_u = \frac{\sum_{\text{tail}} w_{ui}}{\sum w_{ui} + \varepsilon}$$

4. Dual Quota

Dynamic Allocation

$K_{\text{tail}} = \lfloor \alpha_u K \rfloor$

$K_{\text{head}} = K - K_{\text{tail}}$

Candidate Retrieval:

$M_{\text{pop}} \rightarrow \mathcal{C}_{\text{head}} \setminus B_u$

$M_{\text{tail}} \rightarrow \mathcal{C}_{\text{tail}} \setminus B_u$

5. Top-K Merging

Calibrated Merge

Top $K_{\text{head}} \in \mathcal{C}_{\text{head}}$

Top $K_{\text{tail}} \in \mathcal{C}_{\text{tail}}$

Seen-Item Masking

Zero-K Protection

Final Output:

$\mathcal{R}_u = [i_1 \ldots i_K]$